INCH-POUND

MIL-DTL-83723/88E
w/AMENDMENT 1
12 March 2024
SUPERSEDING
MIL-DTL-83723/88E
1 October 2010

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (FLANGE MOUNT, THREADED COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, J, L, N, P AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

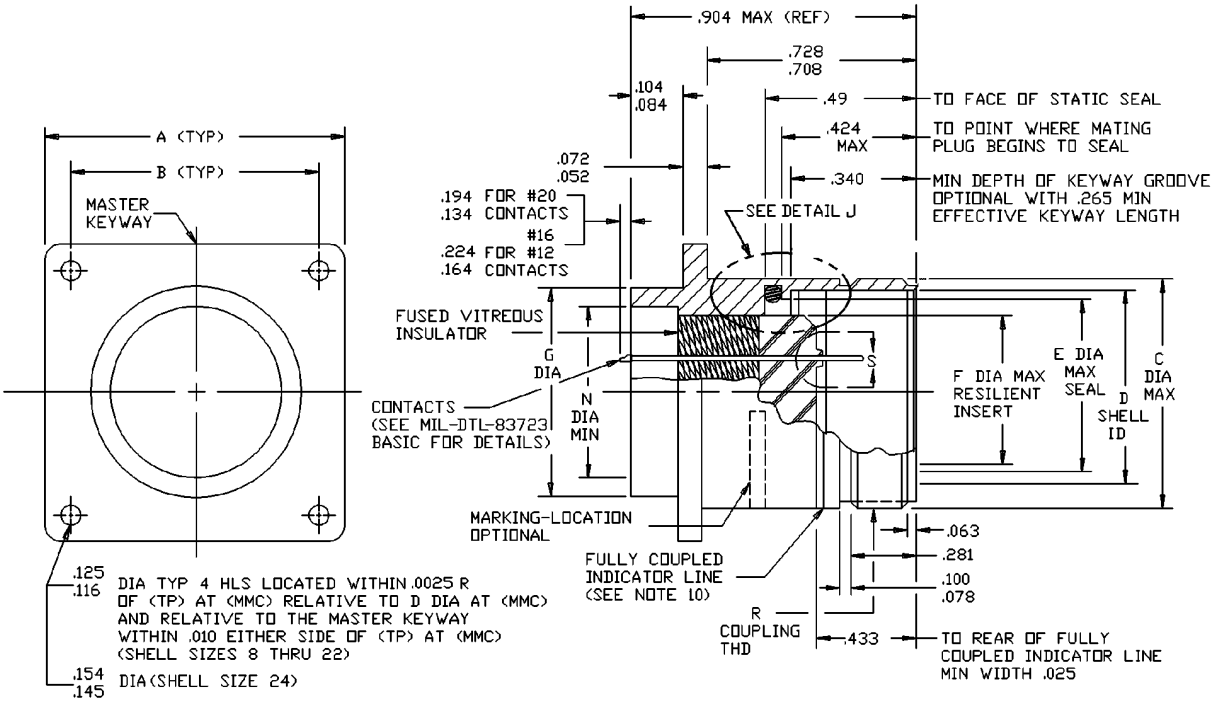


FIGURE 1. Dimensions and configurations.

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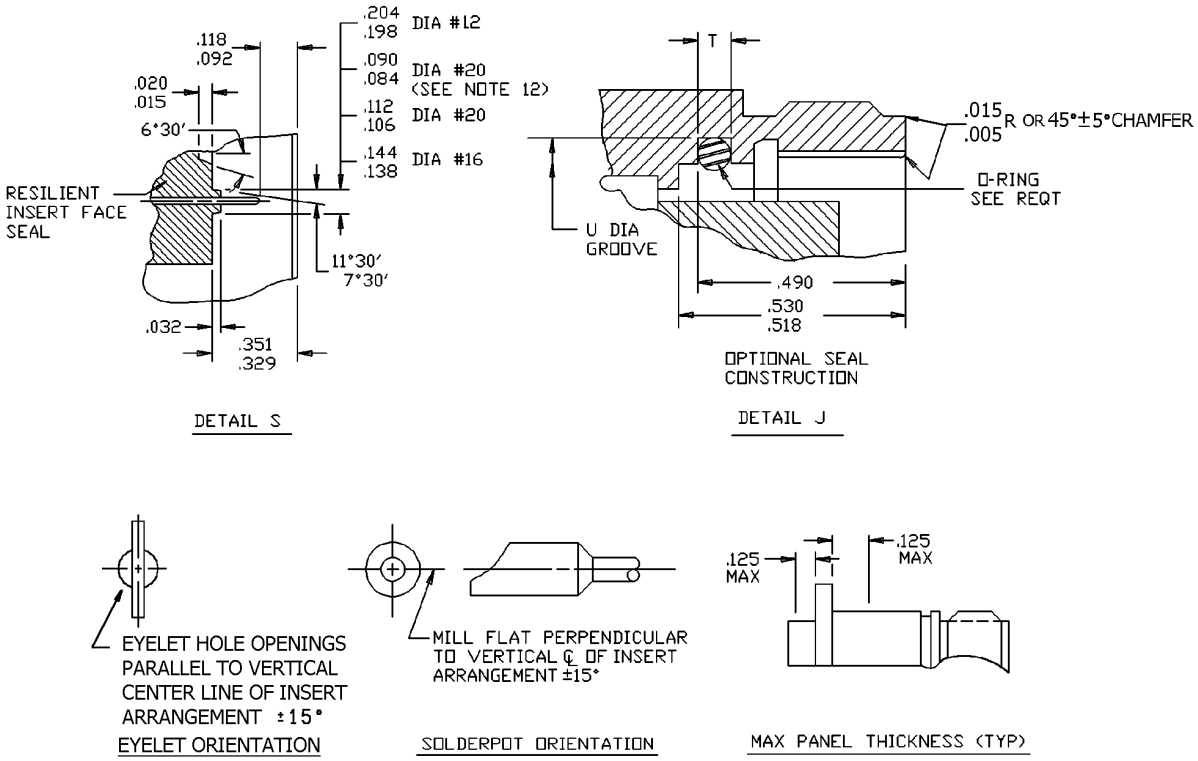


FIGURE 1. Dimensions and configurations – Continued.

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Shell size	A	B	C	D	E	F	G
08	.812	.594	.562	.418 .413	.394	.290	.500 .494
10	.937	.719	.696	.535 .530	.515	.388	.562 .556
12	1.031	.812	.875	.705 .700	.685	.558	.750 .744
14	1.125	.906	.936	.774 .769	.754	.627	.812 .806
16	1.250	.969	1.062	.901 .896	.881	.772	.937 .931
18	1.343	1.062	1.187	1.007 1.002	.987	.860	1.062 1.056
20	1.437	1.156	1.312	1.132 1.127	1.112	.985	1.187 1.181
22	1.562	1.250	1.437	1.257 1.252	1.237	1.110	1.312 1.306
24	1.703	1.375	1.562	1.382 1.377	1.362	1.235	1.437 1.431

FIGURE 1. Dimensions and configurations – Continued.

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Shell size	N	Panel cutout dia	R thread UNEF-2A	T	U +. 005 - .000
08	.400	.572	.562-24	.076 .071	.490
10	.411	.706	.688-24	.089 .084	.636
12	.581	.885	.875-20	.089 .084	.806
14	.650	.946	.938-20	.089 .084	.875
16	.777	1.072	1.062-18	.089 .084	1.002
18	.864	1.197	1.188-18	.089 .084	1.108
20	1.034	1.322	1.312-18	.089 .084	1.233
22	1.133	1.447	1.438-18	.089 .084	1.358

NOTES: (Notes 4 through 9 applicable to contacts only).

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 inches (0.13 mm) for three place decimals, ±.01 inches (0.25mm) for two place decimals and ±1° for all angles.
4. Contact finish: Gold plate .000050 inch (0.00127 mm) minimum in accordance with ASTM B488. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
5. Dimensions apply after plating.
6. Surface finish: 63 ($\sqrt{63}$) all over in accordance with ASME-B46.1, unless otherwise specified.
7. Remove all burrs and sharp edges .003 to .005 inches (0.076 to 0.127 mm) R.
8. Diameters to be concentric within .004 inches (0.102 mm) TIR.
9. Diameters in the same cross-sectional plane to be concentric with each other .003 inches (0.076 mm) TIR.
10. Coupling indicator line shall be color coded red.
11. Shell size 8 will not mate with MIL-DTL-26500 connectors.
12. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

This connector mates with MIL-DTL-83723/86, /91, /95 and /97.

Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangement	Contact cavities
08	03, 98	1, 2, 3
10	06	1, 2, 3, 4, 5, 6

Connector accessories: See SAE-AS85049.

O-ring: See MIL-DTL-83723.

Part or Identifying Number (PIN) example:

M83723/88 H 08 03 N

M83723/	MIL number prefix
88	Specification sheet number
H	Class
08	Shell size
03	Insert arrangement
N	Polarizing position

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-DTL-26500
MIL-DTL-83723/86
MIL-DTL-83723/91
MIL-DTL-83723/95
MIL-DTL-83723/97
MIL-STD-1554
ASME-B46.1
ASTM B488
SAE-AS85049

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2024-097)

Review activities:
Army - AR, MI
Navy - EC
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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